

The same search queries and filters were applied across IEEE Xplore, Scopus, and the ACM Digital Library. No database-specific modifications were applied to the search terms; the same approved search strategy was used across the three databases.

Research Questions (RQs)

RQ No.	Research Question	Search Queries	Search Field	Years	Language	Document Type	Access
RQ1	What IoT architectures and sensing technologies are most effective for reliable bin fill-level monitoring in smart waste management systems?	("Abstract": "Internet of Things" OR "Abstract": IoT) AND ("Abstract": architecture OR "Abstract": framework OR "Abstract": design OR "Abstract": "system model") AND ("Abstract": fill level AND "Abstract": sens*) AND ("Abstract": Waste)	Abstract	2015-2025	English	Journal articles & conference papers	Open access / full-text accessible
RQ2	Which machine learning or time-series models have achieved the highest accuracy in predicting bin fill levels or waste generation trends?	("Abstract": Machine Learning) AND ("Abstract": "Bin Fill Level" OR "Abstract": "Waste") AND ("Abstract": Predict* OR "Abstract": Forecast*)	Abstract	2015-2025	English	Journal articles & conference papers	Open access / full-text accessible

RQ3	What optimization methods are most effective for dynamic waste collection routing, and how are forecasting results integrated into route planning?	("Abstract": "Waste Collection" OR "Abstract": "Solid Waste Management" OR "Abstract": "Smart Waste" OR "Abstract": "Municipal Waste") AND ("Abstract": "route*") AND ("Abstract": "Forecast*" OR "Abstract": "Predict*" OR "Abstract": "Waste Generation" OR "Abstract": "Data-Driven") AND ("Abstract": "Optimization" OR "Abstract": "Machine Learning" OR "Abstract": "Artificial Intelligence" OR "Abstract": "Modeling")	Abstract	2015-2025	English	Journal articles & conference papers	Open access / full-text accessible
RQ4	How is digital twin technology used to model and optimize urban waste collection?	("Document Title": "Digital Twin" OR "Document Title": "DT" OR "Document Title": "Virtual Twin") AND ("Document Title": "waste management" OR "Document Title": "municipal waste" OR "Document Title": "solid waste")	Document Title	2015-2025	English	Journal articles & conference papers	Open access / full-text accessible

RQ5	What operational and environmental benefits have been demonstrated by AI- and IoT-based waste management systems?	("Abstract": "Internet of Things" OR "Abstract": IoT) AND ("Abstract": "artificial intelligence" OR "Abstract": "machine learning" OR "Abstract": "deep learning") AND ("Abstract": waste) AND ("Abstract": benefit* OR "Abstract": impact* OR "Abstract": operational OR "Abstract": environmental OR "Abstract": sustainab* OR "Abstract": cost OR "Abstract": fuel OR "Abstract": "travel distance" OR "Abstract": "route length" OR "Abstract": time OR "Abstract": efficien* OR "Abstract": productivity)	Abstract	2015-2025	English	Journal articles & conference papers	Open access / full-text accessible
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